

Technical Approach for SOE 2025 Azure Migration

1 Planning & Modernization

1.1 Discovery and Assessment

- **Inventory & dependency mapping:** Use ASUK's CMDB and Azure Migrate discovery tools to build an inventory of applications and their dependencies.
- **Qualitative analysis:** For each application, record its tech stack, criticality, complexity and testing requirements.
- **Application prioritization:** Together with ASUK, rank applications by strategic importance and risk.
- **Stakeholder & governance alignment:** Identify ASUK stakeholders (architecture, security, change management).

1.2 Architecture and Wave Planning

- **Landing zone design:** Create separate Azure subscriptions and resource groups for Dev, Test/UAT and Production.
- **Golden image creation:** Build a hardened Windows Server 2025 image in Azure Compute Gallery.
- **Repository & package structure:** Follow the recommended repo layout: directories for environments, branches for features and releases.
- **Branching strategy:** Adopt Gitflow: develop features on 'feature/<ticket>' branches, merge into 'develop' when ready.
- **CI/CD pipeline design:** Implement Azure DevOps multi-stage YAML pipelines. Include build (lint, test), deploy and rollback stages.

1.3 Modernization of Infrastructure-as-Code

- **Terraform modules:** Develop reusable modules for VMs, networks and OS upgrades. Update them to use the latest Terraform versions.
- **Parameterization:** Store environment-specific values in variables; keep secrets in Azure Key Vault.
- **Quality gates:** Run 'terraform fmt', 'terraform validate' and 'tflint' in the pipeline. Generate module documentation.

1.4 Security & Compliance Planning

- **Compliance alignment:** Map controls to APRA CPS 231/234. Enforce encryption at rest, RBAC and audit logging.
- **Baseline policies:** Apply Microsoft's Windows Server 2025 security baseline and enforce via Azure Policy.
- **Security review:** Present detailed designs and pipeline code for ASUK's security approval.

2 Pilot Migration

2.1 Pilot Pipeline Development

- **Upgrade pipeline:** Build an Azure DevOps pipeline that snapshots existing OS and data disks via Azure Backup, upgrades the OS and restores the disks.
- **Configuration management:** Use PowerShell DSC or Ansible to enforce GPO, join domain, and install applications.
- **Logging:** Capture upgrade logs in Azure Monitor; configure alerts for failures or extended duration.

2.2 Pilot Testing & Validation

- **Functional testing:** Verify application start-up, login policies, and key workflows.
- **Policy and security checks:** Confirm GPO compliance, Defender for Cloud recommendations, and audit log collection.
- **Performance comparison:** Compare CPU, memory and disk metrics against pre-migration baselines.
- **Rollback verification:** Test the rollback plan by intentionally failing an upgrade and restoring from a snapshot.

2.3 Pilot Review and Improvement

- Conduct a retrospective with ASUK teams. Capture lessons learned; update Terraform modules, scripts and pipeline.

3 Full Rollout

3.1 Wave Planning & Scheduling

- **Move group definition:** Use dependency mappings to group tightly coupled applications. Plan waves based on criticality and complexity.
- **Communication:** Notify users and owners about maintenance windows and expected downtimes; provide rollback instructions.
- **Capacity readiness:** Ensure Azure quotas and network resources are provisioned ahead of each wave.